

Differential Pair Routing

Tight coupling • Length matching • Escape routing • Layer transitions

The deep dive on routing USB / HDMI / Ethernet / DDR / PCIe diff pairs

Audience: Intermediate Engineering Students

Theory • Layout rules • Length tuning • Vias • Common mistakes

Why Differential Signalling

Two wires, one signal — but worth it

Common-mode rejection

Receiver subtracts the two lines → noise cancels.

Lower EMI

Equal & opposite currents
→ net EM field is small.

Higher speeds

Smaller voltage swings,
faster signalling possible.

Standardised

USB, HDMI, Ethernet, PCIe,
DDR — all differential.

Cost

Two traces, two pins
vs one single-ended — pin cost ×2

Layout complexity

Must keep symmetric and
coupled along entire path

Differential vs Single-Ended

Two fundamentally different signalling approaches

Single-ended

- ▶ One signal trace + ground reference
- ▶ Voltage measured to ground
- ▶ Common-mode noise affects signal directly
- ▶ Lower pin count
- ▶ Used for: SPI clock, I²C, low-speed control
- ▶ Below ~50 MHz at typical PCB lengths

Differential

- ▶ Two signal traces (D+ and D-)
- ▶ Voltage measured between the pair
- ▶ Common-mode noise rejected (CMRR > 40 dB)
- ▶ Higher pin count, more PCB area
- ▶ Used for: USB, HDMI, Ethernet, DDR strobes, PCIe
- ▶ Required for any GHz-class signalling

Coupling — Tight vs Loose

The two diff-pair philosophies

- ▶ Tight coupling: gap $\approx$ trace width — diff pair has its OWN reference
- ▶ Loose coupling: gap $> 2\times$ trace width — each trace references the ground plane
- ▶ Standards mostly want TIGHT coupling for high CMRR
- ▶ Trade-off: tight coupling $\rightarrow$ higher Z_{diff} than $2\times$ single-ended Z_0
- ▶ Typical tight pair: 5/5/5 (5 mil trace, 5 mil gap, 5 mil trace) for 100 Ω diff
- ▶ Loose: 5/15/5 — much wider gap; pair behaves more like $2\times$ single-ended

Industry default

USB, HDMI, Ethernet $\rightarrow$ tight coupling.
DDR strobes $\rightarrow$ loose coupling (each strobe references GND plane independently).
PCIe $\rightarrow$ tight to medium.

Rule of thumb

Pair-to-pair spacing $\geq 4\times$ the within-pair gap. Otherwise neighbouring pairs couple $\rightarrow$ crosstalk.

Length Matching

Two traces must arrive at the same time

- ▶ Skew = difference in arrival time between D+ and D-
- ▶ FR-4 propagation delay ≈ 150 ps/inch (6 ps/mm)
- ▶ 5 mil (0.13 mm) length difference ≈ 1 ps skew
- ▶ USB 2.0 limit: ~ 50 ps pair-skew (≈ 8 mm length difference)
- ▶ USB 3.0 limit: ~ 15 ps pair-skew (≈ 2.5 mm)
- ▶ PCIe Gen 3: ~ 5 ps pair-skew (≈ 0.8 mm)
- ▶ DDR4: byte-lane match ≤ 30 ps (≈ 5 mm)
- ▶ Add 'serpentine' wiggles to the SHORT trace to compensate

Serpentine (Length-Tuning) Patterns

How EDA tools add length

- ▶ Trombone (single zigzag) — old style, ringing at high speed
- ▶ Sawtooth — sharp 90° corners — bad (mostly obsolete)
- ▶ Accordion / meander — gentle 45° — modern default
- ▶ Each wiggle adds ($2 \times$ peak height) to the trace length
- ▶ Spacing between meander turns $\geq 3 \times$ line width
- ▶ Avoid placing meanders close to driver / receiver

KiCad workflow

Select trace → Route → 'Tune length'
or right-click → 'Tune Single Trace Length' / 'Tune Differential Pair Length'.
Specify target length; KiCad auto-meanders.

Don't over-tune

Meanders create small impedance discontinuities. Tune as little as needed. For long traces, place serpentine in the middle, not at endpoints.

Symmetric Routing Rules

Both traces follow the same physical path

- ▶ Both traces on the SAME LAYER (no top/bottom split)
- ▶ Same number of vias for each trace in the pair
- ▶ Both traces turn at the same angle simultaneously
- ▶ When the pair turns, maintain pair spacing through the turn
- ▶ Match number of 45° corners on each trace
- ▶ Pad transitions (escape from BGA / connector): both traces escape symmetrically
- ▶ Bend together — don't have one trace U-turn while the other goes straight
- ▶ If you must traverse a complex region, keep length-match in mind during routing

Escape Routing from BGA / Connectors

Getting diff pairs OUT of dense pin clusters

- ▶ BGA escape: trace fans out from each ball to the nearest free routing area
- ▶ For diff pair pins (e.g., USB on a USB-C BGA controller): escape both pins together
- ▶ Use vias to drop diff pairs onto an inner layer where there's more routing room
- ▶ On the inner layer, maintain controlled-Z + tight coupling
- ▶ Connector pin assignment matters — request diff pairs on adjacent pins
- ▶ Common error: D+ on pin 3, D– on pin 19 → 50 mm asymmetric escape → unfixable skew
- ▶ Pre-fab DFM check: 'do my diff pairs escape symmetrically?'

Via Transitions for Diff Pairs

Vias = impedance discontinuity — minimize and optimize

- ▶ Each via adds ~1 nH inductance + ~0.3 pF capacitance — local impedance bump
- ▶ Diff pair via crossing: BOTH traces get a via (matched discontinuity)
- ▶ Place GND stitching vias adjacent to signal vias (current return path)
- ▶ Back-drill long via stubs for very high speed (PCIe Gen 4+)
- ▶ Microvias / blind vias for fine-pitch BGAs
- ▶ Anti-pad size affects via impedance — larger anti-pad = closer to 50 Ω
- ▶ Use 'via-in-pad' on BGA when escape routing is tight
- ▶ Stitching via spacing: $< \lambda/20$ of the highest frequency (e.g. 5 mm at 5 GHz)

Impedance Discontinuities

Where Z_0 breaks along the trace

- ▶ Width transitions (chip pad → trace width): use tapered transition over 2-3× trace width
- ▶ Connector landing pads: pads are wider than trace → local capacitance
- ▶ Vias: as covered, local bumps
- ▶ Plane splits: signal crosses gap → return current detours → impedance jumps
- ▶ Component pad transitions: avoid placing components in the middle of high-speed traces
- ▶ Length-tuning meanders: small reflections at each wiggle
- ▶ Layer transitions through vias: change in dielectric environment → impedance change
- ▶ Goal: minimise count of discontinuities, AC-couple where the discontinuity is unavoidable

DDR Diff Pair Rules

DQS (data strobe) is the diff pair in DDR

- ▶ DQS / DQS_b = differential data strobe (per byte lane)
- ▶ DDR3/4 byte lane: 8× DQ (single-ended) + 1× DQS (diff) per byte
- ▶ Length matching within byte lane: DQ matched to DQS within 30 ps (~5 mm)
- ▶ DQS diff pair: pair-skew < 5 ps (length match within 0.8 mm)
- ▶ Topology: fly-by for ADDR / CMD / CTRL; T-branch alternatives
- ▶ Reference: VDDQ (DDR4) — drives DQ voltage swing; clean rail mandatory
- ▶ Layer constraint: all DDR signals on layer with continuous GND plane reference
- ▶ Vendor design guides (Micron, Hynix, Samsung) provide exact rules

How to Verify Your Diff Pair

Measurement tools + techniques

TDR (Time Domain Reflectometry)

Measures impedance vs time/distance.
Picks up discontinuities along the trace.

VNA (Vector Network Analyzer)

Measures S-parameters.
Eye diagrams for high-speed validation.

Eye diagram

Visual signal quality.
Receiver-aligned bit-stream overlap.

Skew measurement

Two channels of scope.
Measure delta-t at receiver pins.

HyperLynx / Sigrity SI sim

Pre-layout simulation.
Predict impedance + eye margin.

Pre-compliance scan

Vendor SI suite.
Validate against USB / HDMI / PCIe specs.

Common Diff Pair Mistakes

What scope captures reveal

- ▶ Diff pair traces on DIFFERENT layers — impedance varies, no CMRR
- ▶ Long unmatched escape from BGA — pair-skew exceeds budget
- ▶ Plane splits crossed by diff pair — return discontinuity → EMI
- ▶ Vias on only one trace of the pair → asymmetric loading
- ▶ Tight coupling broken at width transitions (connector pads, vias)
- ▶ Pair-to-pair spacing too small → crosstalk between USB and Ethernet
- ▶ Stub at receiver pad (extra branch) — reflections
- ▶ Forgetting to mark the pair in the EDA tool — auto-router won't route diffly

Best Practices

Twelve habits for high-yield diff pair design

- ▶ Mark diff pairs in schematic with net classes
- ▶ Choose controlled-Z stackup early
- ▶ Tight coupling throughout (width = gap)
- ▶ Both traces on same layer
- ▶ Symmetric escape from connectors
- ▶ Equal number of vias on both legs
- ▶ Continuous reference plane (no splits)
- ▶ Length-tune at midpoint (not at endpoints)
- ▶ GND stitching vias near layer changes
- ▶ Avoid stubs at receiver
- ▶ Verify with TDR / VNA after fab
- ▶ Document spec in fab quote (controlled Z + pair skew)

Key Takeaways

Key takeaways from this lecture

Tight coupling = CMRR

Gap $\approx$ trace width.

Length match = timing

Pair-skew matters more at higher speeds.

Symmetric routing

Both traces, same path.

Reference plane sacred

Never cross plane splits.

Vias matter

Match transitions; stitch ground.

Verify with TDR + eye

Measure, don't assume.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- ▶ Eric Bogatin — 'Signal & Power Integrity Simplified'
- ▶ Stephen Hall — 'Advanced Signal Integrity for High-Speed Digital Designs'
- ▶ Howard Johnson — 'High-Speed Digital Design'
- ▶ KiCad differential pair tuning tutorial (docs.kicad.org)
- ▶ Robert Feranec YouTube — diff pair walkthroughs
- ▶ USB-IF / HDMI Forum / PCI-SIG compliance documents
- ▶ Micron / Samsung / Hynix DDR layout design guides
- ▶ Polar Si9000 — controlled impedance + diff pair calculations

Thank you • Build something this week.